

Experimental Roadmap for Validating Fenchel–Young Surrogates in Kidney Exchange

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1 Motivation

The purpose of the experiments is not only to compare decision-focused learning (DFL) against a two-stage predict-then-optimize pipeline, but also to establish whether the Fenchel–Young (FY) perturbed-optimizer loss [Berthet et al., 2020] is a meaningful surrogate for the true decision loss in our kidney exchange setting.

This distinction matters because the realistic synthetic benchmark used in our main experiments has a complex reward mechanism. The ground-truth edge reward is not a simple linear function of utility, but a nonlinear product of transplant success probability, QALY, scarcity priority, graph-structural variables, and deterministic edge-level noise. Therefore, a direct comparison between DFL and a two-stage baseline on this benchmark alone cannot isolate the behavior of the surrogate loss. If DFL improves the final matching score, the improvement may be due to the surrogate, the choice of perturbation scale, the warm start, or model capacity. If DFL does not improve, the failure may instead be caused by model misspecification or an overly complex label-generation process.

We therefore separate the experimental study into several layers, moving from a controlled noisy clean-linear surrogate-validity test to the full realistic synthetic benchmark.

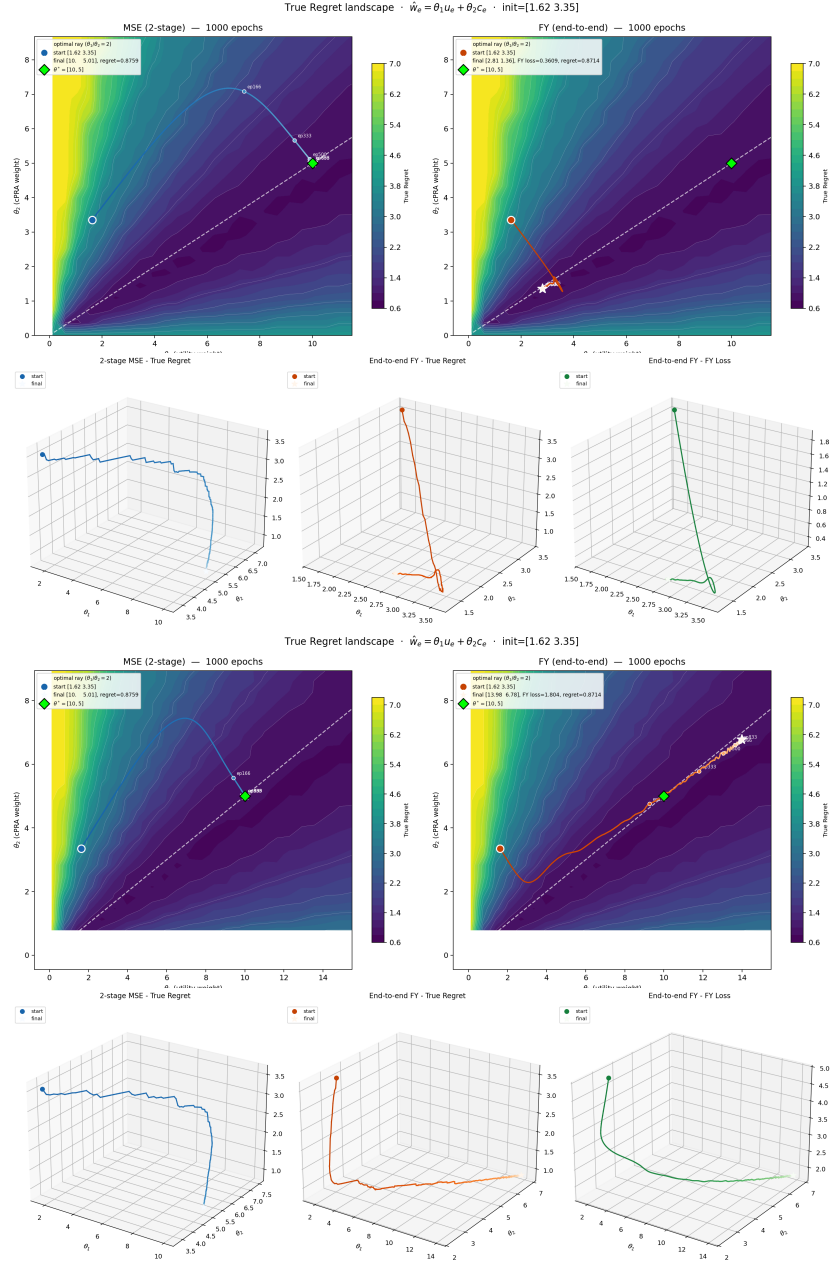


Figure 1: Step 1 diagnostics under the current noisy clean-linear label. The four panels, from top to bottom, show the $\epsilon = 0.2$ true-regret landscape, $\epsilon = 0.2$ trajectory metrics, $\epsilon = 1.0$ true-regret landscape, and $\epsilon = 1.0$ trajectory metrics. All panels use the same 100 graph instances from dataset/processed/step1noisy_linear_sigma010.

2 Experimental Roadmap

2.1 Step 1: Surrogate Validity Under a Noisy Clean Linear Label

The first experiment should validate whether the FY loss is directionally aligned with the true regret in a controlled setting. The current Step 1 run uses a noisy clean-linear label:

$$w_e^{\text{true}} = \max \{0, (10u_e + 5c_e)(1 + \xi_e)\},$$

where u_e is the edge utility and c_e is the recipient cPRA associated with edge e . The deterministic edge-level noise ξ_e is generated by the processed batch `dataset/processed/step1_noisy_linear_sigma010`, with multiplicative noise scale $\sigma = 0.10$. The corresponding prediction model remains the two-feature linear probe

$$\hat{w}_e(\theta) = \theta_1 u_e + \theta_2 c_e.$$

This setting preserves a transparent two-feature signal while adding controlled edge-level variation that the two-feature probe cannot perfectly fit. If the FY surrogate is appropriate, its trajectory should still move toward low-regret regions of the true decision landscape, even though exact reward recovery is no longer possible.

The main diagnostic objects are:

- the true regret landscape $R(\theta_1, \theta_2)$;
- the FY surrogate landscape $L_{\text{FY}}(\theta_1, \theta_2)$;
- the two-stage MSE solution θ_{2s}^* ;
- the true oracle set

$$\Theta^* = \arg \min_{\theta} R(\theta);$$

- the FY minimizer

$$\theta_{\text{FY}}^*(\epsilon) = \arg \min_{\theta} L_{\text{FY}, \epsilon}(\theta).$$

The primary metric should be regret gap rather than parameter distance:

$$\Delta R(\epsilon) = R(\theta_{\text{FY}}^*(\epsilon)) - \min_{\theta} R(\theta).$$

This is important because the KEP solver is piecewise constant, so the true regret landscape often has an optimal plateau rather than a unique isolated minimizer. In such cases, parameter-space distance to one selected oracle point is not well-defined, whereas ΔR remains meaningful.

This step answers the question:

Does the FY surrogate point toward the true low-regret region when the predictive model captures the main clean signal but cannot represent deterministic edge-level noise?

In the current Step 1 diagnostic, this question is tested on 100 graph instances, using the same sampled instances for the MSE and FY trajectories. We track the full optimization trajectories of both the two-stage MSE objective and the end-to-end FY objective in the (θ_1, θ_2) plane, and evaluate each point by true regret. The saved trajectories contain 1001 points (1000 training epochs plus the initialization). The initial parameter has average true regret 3.5402.

The two-stage MSE trajectory still recovers the clean component of the noisy label, converging to approximately $(10.00, 5.01)$. However, because of the deterministic multiplicative noise, the final MSE solution does not reach zero true regret; its final average true regret is 0.8759. The FY trajectories show a different, decision-focused behavior. They do not need to recover the clean parameter vector $(10, 5)$. Instead, they move through regions that improve the induced KEP decisions under the noisy label. For $\epsilon = 0.2$, the trajectory reaches its best observed true regret early, at epoch 42, before continuing to reduce the FY loss. For $\epsilon = 1.0$, the trajectory grows to a larger parameter scale and reaches its best observed true regret later, around epoch 781.

method	ϵ	final θ	final loss/regret	best regret
MSE	–	(10.0017, 5.0147)	$R = 0.8759$	0.8759
FY	0.2	(2.8135, 1.3589)	$L_{\text{FY}} = 0.3609$, $R = 0.8714$	0.7183 (ep. 42)
FY	1.0	(13.9796, 6.7825)	$L_{\text{FY}} = 1.8042$, $R = 0.8714$	0.8668 (ep. 781)

These results support the intended interpretation of the diagnostic. The two-stage method identifies the clean linear component of the noisy label, while the FY surrogate searches in a decision-induced geometry. In this noisy controlled setting, the FY trajectory can visit lower-regret regions than the MSE endpoint, but the final FY endpoint is not necessarily the best-regret iterate. This distinction is important for interpreting FY optimization: the surrogate loss may continue to optimize perturbation robustness even after the deterministic KEP decision has already reached a low-regret region.

This leads to the following practical observations.

1. **The FY loss is decision-oriented rather than reward-fitting.** Unlike the two-stage MSE trajectory, which recovers the clean component $(10, 5)$, the FY trajectory moves according to changes in the induced KEP decision. With $\epsilon = 0.2$, the best observed FY point has true regret 0.7183, below the final MSE regret 0.8759, despite being far from $(10, 5)$ in parameter space.
2. **The perturbation scale controls decision stability and parameter scale.** With $\epsilon = 0.2$, the final FY endpoint is $(2.8135, 1.3589)$. With $\epsilon = 1.0$, the final endpoint is $(13.9796, 6.7825)$. The larger perturbation scale encourages a larger prediction scale, because the same decision must remain stable under stronger perturbations. This is consistent with the FY gradient form

$$\nabla_{\hat{w}} L_{\text{FY}}(\hat{w}; y^*) = \mathbb{E}_z[y^*(\hat{w} + z)] - y^*,$$

which penalizes deviations between the perturbation-averaged optimizer output and the true-weight optimal decision.

3. **The best-regret point need not be the final FY endpoint.** For $\epsilon = 0.2$, the minimum observed true regret is 0.7183 at epoch 42, while the final true regret is 0.8714. For $\epsilon = 1.0$, the best observed true regret is 0.8668 at epoch 781, and the final true regret is 0.8714. This shows that in the noisy setting, early stopping or model selection by true regret can matter.
4. **Zero regret is unrealistic for this two-feature probe.** The MSE endpoint is close to $(10, 5)$, but its final regret remains 0.8759. Thus the current Step 1 is no longer a pure “recover-the-clean-parameter” sanity check; it is a controlled decision-quality diagnostic under mild label noise.

2.2 Step 2: Controlled Model Misspecification

The second experiment should introduce a limited amount of misspecification. For example, the true reward may include an additional feature or interaction:

$$w_e^{\text{true}} = au_e + bc_e + d \text{age}_e^{\text{donor}},$$

or

$$w_e^{\text{true}} = au_e + bc_e + d u_e c_e.$$

The prediction model can then be kept deliberately simpler:

$$\hat{w}_e(\theta) = \theta_1 u_e + \theta_2 c_e.$$

This setting tests whether FY optimization is robust when the model cannot perfectly recover the edge rewards. The question is no longer whether FY can recover the true label function, but whether it can still produce better decisions than an MSE-trained predictor.

The same landscape diagnostics should be reported:

$$R(\theta), \quad L_{\text{FY}, \epsilon}(\theta), \quad R(\theta_{2s}^*), \quad R(\theta_{\text{FY}}^*), \quad \Delta R.$$

This step answers the question:

When the predictive model is misspecified, does the FY surrogate still move the parameters toward lower true regret than the two-stage MSE solution?

2.3 Step 3: Training-Level DFL vs Two-Stage Comparison

After the landscape-level surrogate diagnostics, we should run actual training experiments on the same controlled settings. Both methods should use the same train/validation/test split, the same model class, and the same KEP solver.

The two-stage baseline first minimizes prediction error:

$$\min_{\theta} \sum_e (\hat{w}_e(\theta) - w_e^{\text{true}})^2,$$

then solves the KEP using the predicted weights $\hat{\mathbf{w}}(\theta)$.

The DFL model starts either from the two-stage checkpoint or from random initialization and then updates the predictor using the FY gradient estimate:

$$\nabla_{\hat{\mathbf{w}}} L_{\text{FY}} \approx \bar{y}_\epsilon(\hat{\mathbf{w}}) - y^*,$$

where

$$\bar{y}_\epsilon(\hat{\mathbf{w}}) = \frac{1}{M} \sum_{m=1}^M S(\hat{\mathbf{w}} + \epsilon z^{(m)}), \quad z^{(m)} \sim \mathcal{N}(0, I).$$

The primary evaluation metric should be final achieved ground-truth score, or equivalently average regret:

$$R(\theta) = \frac{1}{N} \sum_{i=1}^N \left[(\mathbf{w}^i)^\top y^*(\mathbf{w}^i) - (\mathbf{w}^i)^\top S(\hat{\mathbf{w}}^i(\theta)) \right].$$

This step answers the question:

Does the landscape-level alignment between FY and true regret translate into a training-level improvement over the two-stage baseline?

2.4 Step 4: Realistic Synthetic Benchmark

The final main benchmark should use the currently processed dataset, now named `dataset/processed/realistic_synthetic_benchmark`. This benchmark uses the nonlinear medical reward:

$$w_e^{\text{true}} = E[\text{transplant}]_e \cdot \text{QALY}_e \cdot \text{priority}_e \cdot (1 + \xi_e).$$

This label incorporates utility, recipient cPRA, donor and recipient age, blood type context, arc failure probability, vertex failure probability, scarcity, donor flexibility, reciprocal edges, and deterministic edge-level noise.

This benchmark should be described as a realistic synthetic benchmark rather than a clean surrogate-identification experiment. Its role is to test whether the DFL pipeline improves exchange-plan quality in a richer synthetic environment.

The interpretation should be careful:

- If DFL improves over two-stage, this provides empirical evidence that the FY-based pipeline can help in a complex KEP setting.
- If DFL does not improve, this does not by itself prove that the FY surrogate is invalid; the failure may come from model misspecification, insufficient features, perturbation scale, or optimization instability.

This step answers the question:

Under a medically motivated synthetic reward model, does FY-based DFL improve the final kidney exchange objective compared with the two-stage pipeline?

2.5 Step 5: Sensitivity and Ablation Studies

The last layer should quantify which design choices affect the result. The most important ablations are:

- perturbation scale ϵ ;
- number of perturbation samples M ;
- warm start from the two-stage solution versus random initialization;
- feature set: utility only, utility+cPRA, small linear features, failure context features, and full features;
- label noise on versus off;
- solver formulation, such as CF, PIEF, and hybrid.

For the perturbation scale, it is important to distinguish absolute and relative noise. Since the KEP solver is an argmax, its output is invariant to positive rescaling of all weights. Therefore the effective smoothing strength is governed by the relative scale of the perturbation:

$$S(\hat{\mathbf{w}} + \epsilon z) = S\left(\frac{\hat{\mathbf{w}}}{\epsilon} + z\right).$$

Thus, ϵ should be interpreted relative to the scale of $\hat{\mathbf{w}}$. In the landscape plots, ϵ may be calibrated relative to the MSE weight scale, while in end-to-end training it may be used as an absolute noise standard deviation. This difference should be reported explicitly.

3 Recommended Presentation Order

The final paper should present the experiments in the following order:

1. noisy clean-linear surrogate-validity benchmark;
2. controlled misspecification benchmark;
3. training-level DFL versus two-stage comparison on controlled labels;
4. realistic synthetic benchmark;
5. epsilon, sample-size, feature-set, and initialization ablations.

This ordering separates mechanism validation from final empirical performance. It allows the paper to make precise claims: first, whether the FY surrogate is aligned with true regret under controlled conditions; second, whether this alignment survives moderate misspecification; and finally, whether it yields practical improvements on the realistic synthetic KEP benchmark.

References

Quentin Berthet, Mathieu Blondel, Olivier Teboul, Marco Cuturi, Jean-Philippe Vert, and Francis Bach. Learning with differentiable perturbed optimizers. In *Advances in Neural Information Processing Systems*, volume 33, pages 9508–9519, 2020.